



Practical No 5

Aim

Designing a Todo-list using Typescript

Apparatus / Software Required

- Visual Studio Code / Any Code Editor
- TypeScript
- Node.js
- Web Browser

Theory

A To-Do List is a simple web application used to manage and organize tasks. It allows users to add new tasks, view the existing tasks, mark tasks as completed, and delete tasks when they are no longer required.

In this practical, TypeScript is used to develop the To-Do List application. TypeScript provides static typing and better code organization compared to JavaScript.

The application accepts a task from the user, stores the task in a collection, and displays the task dynamically on the webpage. The user can interact with each task using buttons such as **Add**, **Complete**, and **Delete**.

Concepts Required

1. TypeScript

TypeScript is a superset of JavaScript developed by Microsoft. It adds static typing to JavaScript and helps developers write reliable, readable, and maintainable code.

Features:

- Static Type Checking
- Object-Oriented Programming Support
- Better Error Detection
- Improved Code Readability
- Easy Maintenance

Example:

```
let task: string = "Complete Assignment";  
let completed: boolean = false;
```

2. Variables and Data Types

Variables are used to store information related to tasks.

Common data types used in a To-Do List are:

- string – stores the task description
- number – stores the task ID
- boolean – stores the completion status



Example:

```
let taskName: string = "Study TypeScript";  
let taskId: number = 1;  
let isCompleted: boolean = false;
```

3.Functions

Functions are used to perform specific operations in the To-Do List.

The application can use functions for:

- Adding a task
- Displaying tasks
- Completing a task
- Deleting a task

Example:

```
function addTask(title: string): void {  
    console.log("Task Added: " + title);  
}
```

4.Arrow Functions

Arrow functions provide a shorter syntax for defining functions in TypeScript.

Example:

```
const deleteTask = (id: number): void => {  
    console.log("Deleting task: " + id);  
};
```

Arrow functions are useful when working with event handlers and array methods.

5. Array Methods

Array methods are used to perform operations on the list of tasks.

Commonly used methods include:

- push() – adds a new task
- forEach() – processes each task
- filter() – removes or selects tasks

Example:

```
tasks = tasks.filter((task) => task.id !== id);
```

The above statement removes the task whose ID matches the given ID.

6. DOM (Document Object Model)

The **Document Object Model (DOM)** represents an HTML document as a tree of objects. TypeScript can use the DOM to access, modify, create, and remove HTML elements dynamically.

In a To-Do List application, DOM methods are used to read user input, display tasks, create task elements, and update the webpage.



Common DOM Methods Used:

a) getElementById()

It is used to access an HTML element using its unique ID.

```
const taskInput =  
    document.getElementById("taskInput") as HTMLInputElement;
```

b) querySelector()

It is used to select the first HTML element that matches a CSS selector.

```
const taskList =  
    document.querySelector("#taskList") as HTMLUListElement;
```

c) createElement()

It is used to create a new HTML element dynamically.

```
const li = document.createElement("li");
```

d) appendChild()

It is used to add a newly created element as a child of another element.

```
taskList.appendChild(li);
```

e) remove()

It is used to remove an HTML element from the webpage.

```
li.remove();
```

f) addEventListener()

It is used to respond to user actions such as button clicks.

```
addButton.addEventListener("click", () => {  
    console.log("Task Added");  
});
```

7. Input Validation

Input validation ensures that the user enters a valid task before adding it to the list.

Example:

```
if (taskInput.value.trim() === "") {  
    alert("Please enter a task");  
    return;  
}
```

The trim() method removes extra spaces from the input.

Benefits:

- Prevents empty tasks
- Reduces invalid input
- Improves application reliability
- Enhances user experience



8. Type Assertion

Type assertion tells TypeScript what type of HTML element is being accessed.

Example:

```
const taskInput =
```

```
    document.getElementById("taskInput") as HTMLInputElement;
```

Here, as `HTMLInputElement` tells TypeScript that the selected element is an HTML input element. This allows us to access its value property.

```
let task = taskInput.value;
```

9. Event Handling

Event handling is used to respond to actions performed by the user.

For example, when the user clicks the **Add Task** button, an event is generated and the task is added to the `lis`

`t`.

Example:

```
addButton.addEventListener("click", () => {
```

```
    addTask(taskInput.value);
```

```
});
```

Here, the click event executes the function when the button is clicked.

10. Dynamic HTML Elements

The To-Do List creates and updates HTML elements dynamically using TypeScript.

For example, a new `<li>` element can be created for every task.

Example:

```
const li = document.createElement("li");
```

```
li.textContent = task.title;
```

```
taskList.appendChild(li);
```

This allows newly added tasks to appear immediately on the webpage without manually changing the HTML.

11. Real-World Applications

To-Do List applications are commonly used in real-world applications to manage and organize daily tasks, work activities, and reminders.

Examples include:

- **Personal Task Management** – Managing daily activities such as shopping, exercise, study, and appointments.
- **Student Task Management** – Maintaining assignments, practicals, project work, and examination preparation.
- **Office Task Management** – Managing meetings, reports, deadlines, and assigned work.
- **Project Management** – Tracking development tasks and their completion status.
- **Reminder Applications** – Keeping track of important activities and pending work.



Shri Yashwantrao Bhonsale Education Society's

YASHWANTRAO BHONSALE INSTITUTE OF TECHNOLOGY

(DTE CODE : 3470) (MSBTE Code : 1742)

Approved by AICTE, DTE & Affiliated to Mumbai University & MSBTE Mumbai
(NBA Accredited ME, CE, EE Diploma Programs)

Conclusion:
